CHELSEA OGBONNA

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EDUCATION

Northwestern University — **Evanston, IL**

B.S. Biomedical Engineering & M.S. Mechanical Engineering — *Anticipated Graduation: June 2026*

- **Relevant Courses:** Advanced Mechatronics, Biomedical Signals and Systems, Thermodynamics

EXPERIENCE

Microfluidic Device Design for Syncytial Knot Isolation, *Team Leader* — *Sept 2025 – Present*

- Developing a method to efficiently isolate syncytial knots from placental explant tissue for the Hieshy Lab.
- Designing, testing, and creating a microfluidic system using CAD, COMSOL, and photoresist SUEX.

Rogers Lab, *Research Assistant* — *Jun 2025 – Present*

- Fabricating and testing Transcatheter Aortic Valve Implantation (TAVI)–integrated pacemaker devices designed to provide simultaneous treatment for severe aortic stenosis and cardiac conduction abnormalities.
- 3D printed prototypes and functional parts for bench-top flow testing to validate design performance.
- Support in vivo validation using sheep models to assess device performance in cardiac environments.

Solutions Lab, *Research Assistant* — *Jun 2025 – Present*

- Trained over 150 Evanston Police Department officers on the use of transparency statements, which are verbal techniques where officers clearly communicate their intentions to build trust and improve interactions.
- Coding and analyzing body-worn camera footage to quantify the impact of transparency statements.

Project Commission on Urgent Relief and Equipment (C.U.R.E), *Intern* — *Jun 2025 – Sept 2025*

- Tested, certified, and inventoried over 150 life-saving medical devices for 10+ countries.
- Repaired critical biomedical equipment such as anesthesia machines, surgical beds, and X-ray systems.

Distal Biceps Tendon Repair, *Research Assistant* — *Jun 2024 – Aug 2024*

- Led comprehensive data collection, cleaning, and preprocessing of 50-70 peer-reviewed studies.
- Collaborated with senior researchers to analyze clinical outcomes, extracting key insights from literature.

PROJECTS

Haptic Diabetic Foot Ulcers — *Jan 2026 – Present*

- Designing a wearable gait analysis system using a MPU6050 and force-sensitive resistors to collect biomechanical walking data for early detection of diabetic foot complications.

Biosignal Acquisition and Control Systems: Blood Pressure — *Mar 2025 – Jun 2025*

- Developed a complete blood pressure measurement system through circuit design and signal acquisition.
- Extracted heart rate and calculated systolic, diastolic, and mean arterial pressure from processed data.

Biosignal Acquisition and Control Systems: EKG and EMG Signal Processing — *Jan 2025 – Mar 2025*

- Designed and built 4 analog EMG/EKG circuits, applying amplification and filtering techniques to extract physiological signals while achieving >90% cost savings versus commercial systems.
- Developed a Python-based FFT analysis pipeline for biosignal data using the nLab API, reducing signal analysis time by ~90% and enabling rapid system performance evaluation.

Design Thinking and Communication 1, *Team Leader* — *Jan 2023 – Mar 2023*

- Collaborated with team members to create a recycling center for a condominium in Chicago by designing recycling bins, making a Cardboard Recycler and posters with workshop tools.
- Increased recycling rates by 40% through strategic interventions; utilized SOLIDWORKS and Siemens NX.

LEADERSHIP

National Society of Black Engineers (NSBE), *Academic Excellence Chair* — *Mar 2022 – Mar 2024*

- Led a highly successful study session, uniting 50 students from NSBE, Society of Women Engineers (SWE), Society of Hispanic Engineers (SHPE) and Women in Computing (WIC), fostering cross-club collaboration.
- Increased club meeting attendance by 70% by strategically promoting NSBE's value and enhancing community engagement through personal contacts and social media marketing.

SKILLS

Skills: Python, Matlab, C++, Microsoft Office, 3D Printing, Soldering, APIs, Technical Writing, Handheld Tools
Software: SolidWorks, Siemens NX, Zoom, Slack, Trello